

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

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[Lab.1] Cohesive Energy	 <u>3</u>
[Lab.2] Li diffusion activation barrier in bulk structure	 <u>12</u>
[Lab.3] Adsorption energies of Li atoms for different coverage	 <u>18</u>
[Lab.4] Li diffusion activation barrier on the surface	 <u>42</u>
[Lab.5] Interaction of Li Ion with Ethylene Carbonates	 <u>47</u>

[Lab.5] Interaction of lithium ion with Ethylene Carbonates

Learning Objectives:

- Calculate the binding energy between lithium ions and molecules using density functional theory. Vary the concentration of molecules and analyze the resulting changes in binding energy and electronic structure.

Materials: Li ion-EC molecules



MatSQ Credit: ~ 168 CPU hour (\$ 42)

[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

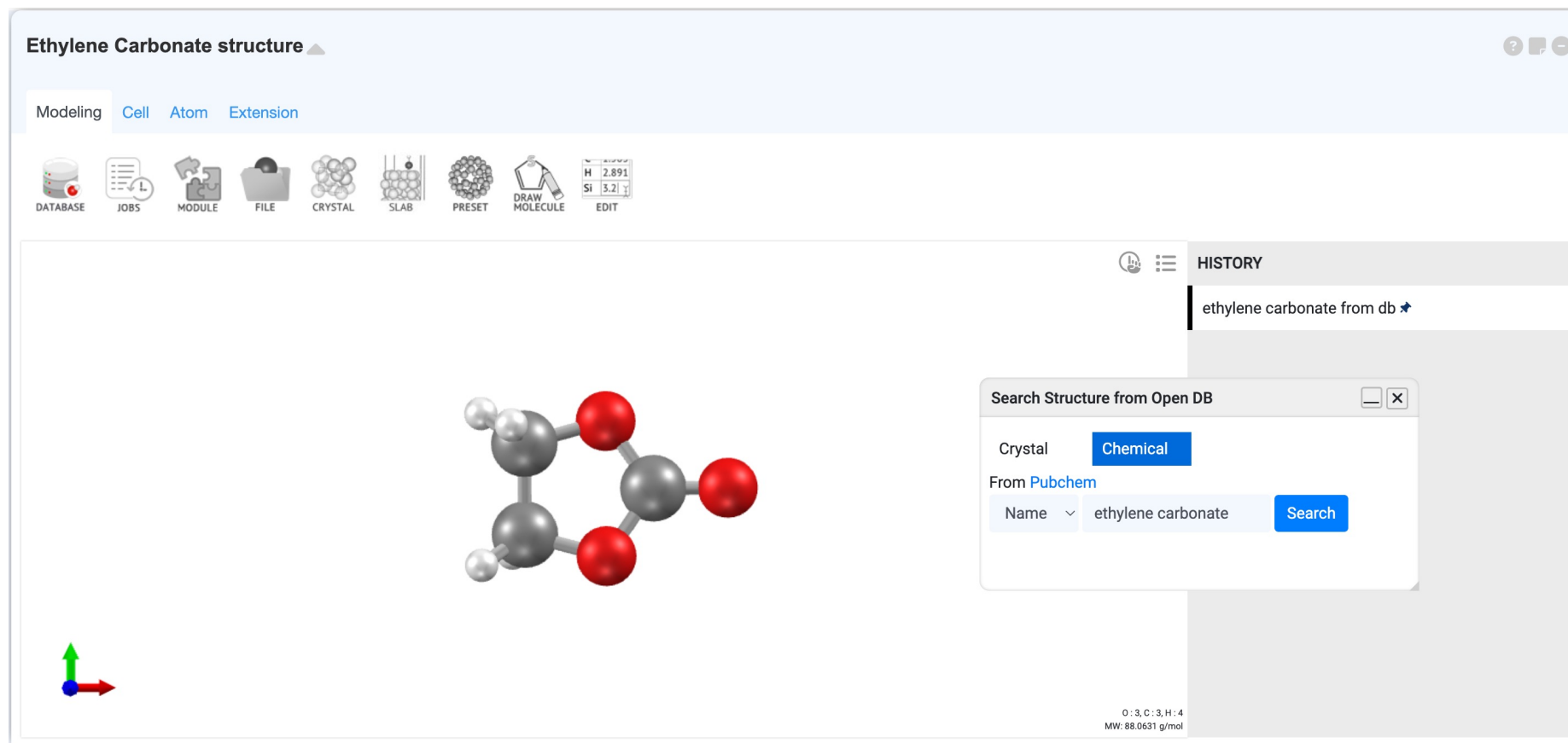
Li⁺-EC

1. Build Ethylene Carbonate (EC)
2. Calculate EC molecules
3. Calculate Li ion energy
4. Build binding structures
5. Calculate atomic structure optimization (Li-ion with n EC molecules)
6. Calculate binding energies

MatSQ Credit: ~ 168 CPU hour (\$ 42)

[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

1. Build Ethylene Carbonate (EC)



[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

2. Calculate optimization of Ethylene Carbonate (EC)

GAMESS   

Gamess.x

Result

Fix atoms ▼

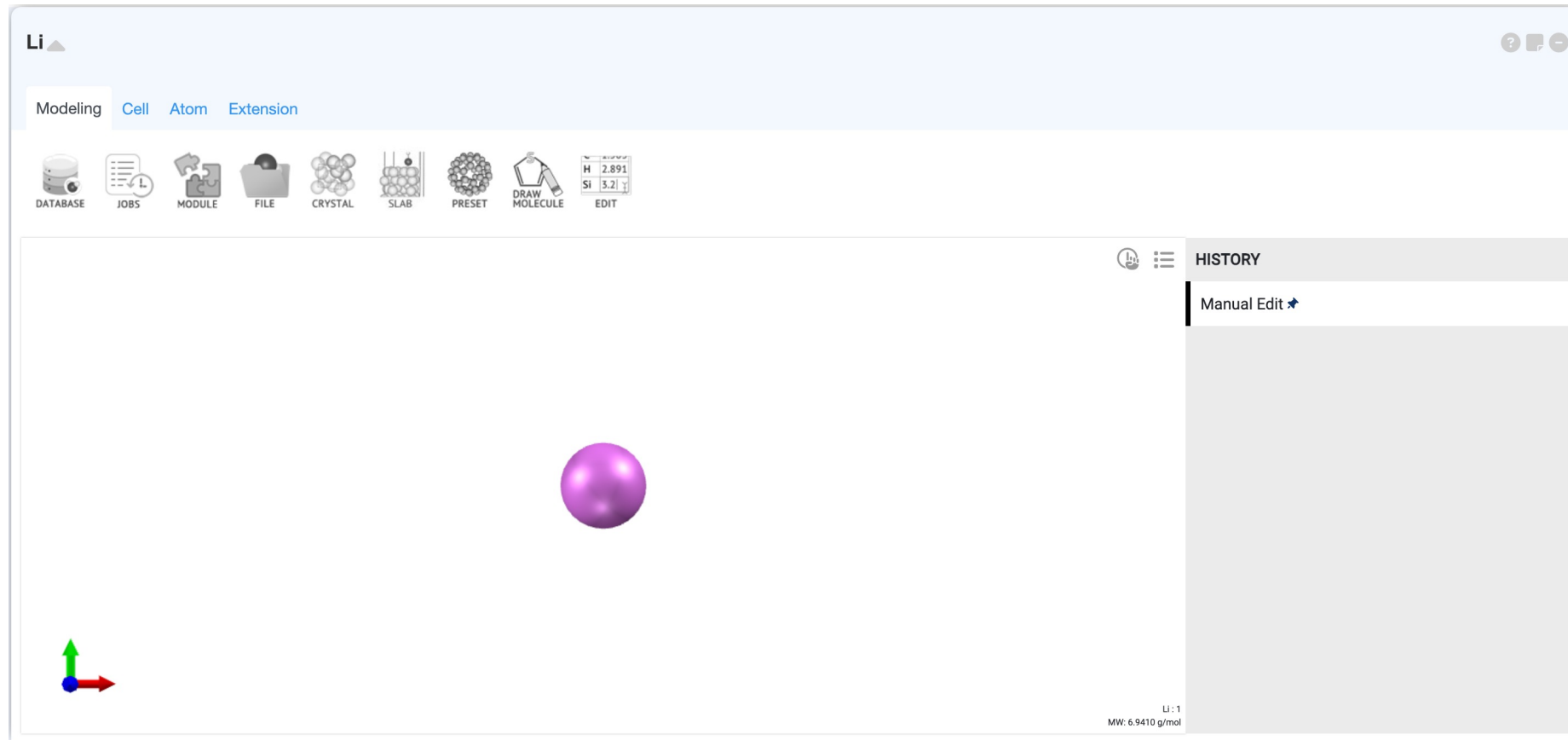
Scripting Option

General ▼

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SCF Type	RHF ▼	DIRSCF	.TRUE. ▼	Initial Orbital	Huckel
MPLEVL	0 ▼	\$BASIS		\$PCM	
RUN Type	OPTIMIZE ▼	BASIS	6-31G** ▼	Solvent effect	NONE ▼
DFT Type	B3LYP ▼	CUSTOM BASIS			
TDDFT	NONE ▼			\$FORCE	
Charge	0			METHOD	SEMINUM ▼
Multiplicity	1			Temperature (K)	298.15
ISPHER	-1 ▼				
PP	NONE ▼				
\$STATPT					
Convergence tolerance (Hartree/Bohr)	0.0001				
Max iteration step	500				
Hessian matrix	GUESS ▼				
IR frequency	.TRUE. ▼				

[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

3. Calculate Li ion energy



[Lab.5] Interaction of lithium ion with Ethylene Carbonates

3. Calculate Li ion energy

GAMESS   

Gamess.x

Result

Fix atoms ▼

Scripting Option

General ▼

\$CONTROL		\$SCF		\$GUESS	
SCF Type	RHF ▼	DIRSCF	.TRUE. ▼	Initial Orbital	Huckel
MPLEVL	0 ▼	\$BASIS		\$PCM	
RUN Type	ENERGY ▼	BASIS	6-31G** ▼	Solvent effect	NONE ▼
DFT Type	B3LYP ▼	CUSTOM BASIS			
TDDFT	NONE ▼				
Charge	1				
Multiplicity	1				
ISPHER	-1 ▼				
PP	NONE ▼				

Input Script

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$GUESS GUESS=HUCKEL $END
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Job Submit

Resource :

Spot ▼

PPN :

2 ▼

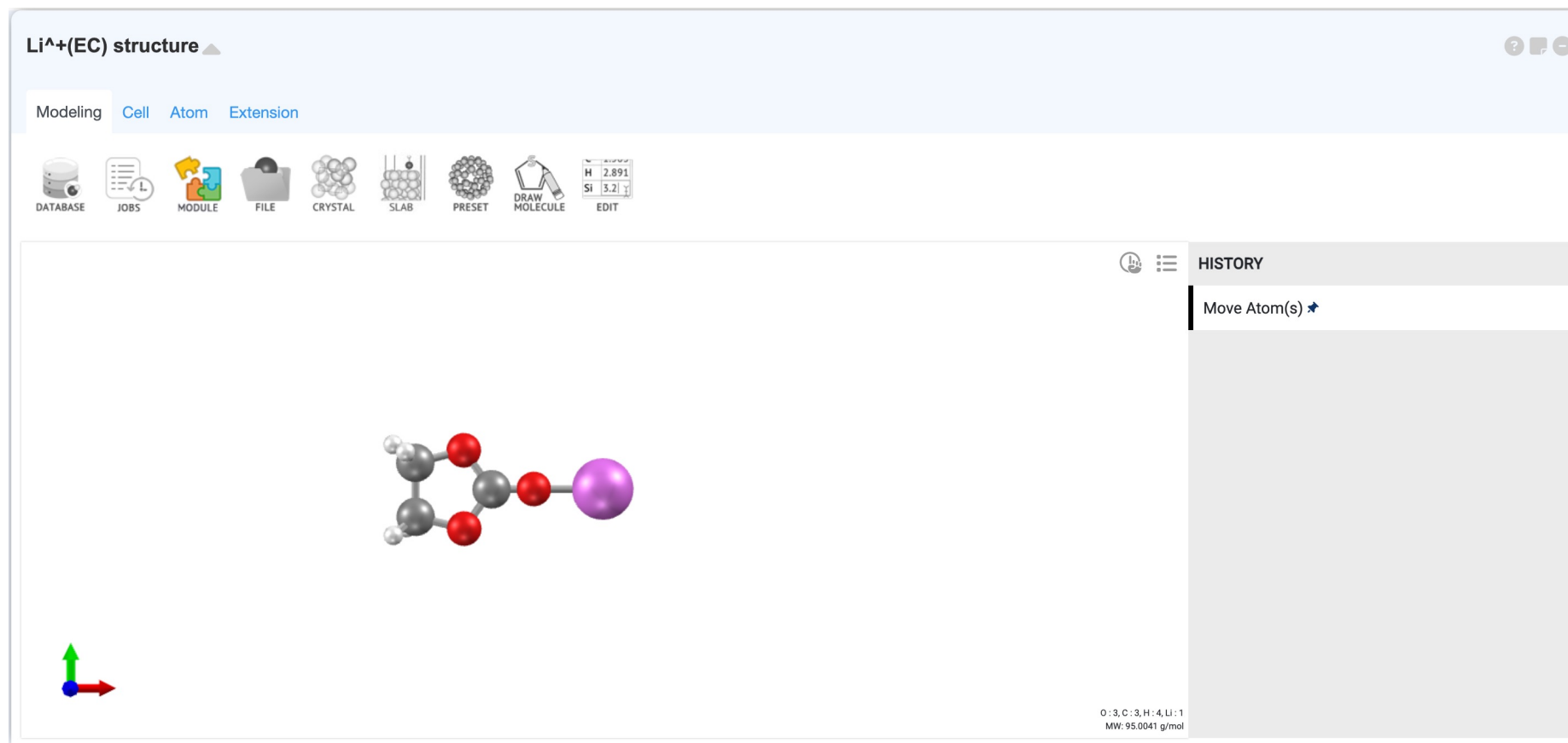
 ram : 8 GB

Node type :

Computing ▼

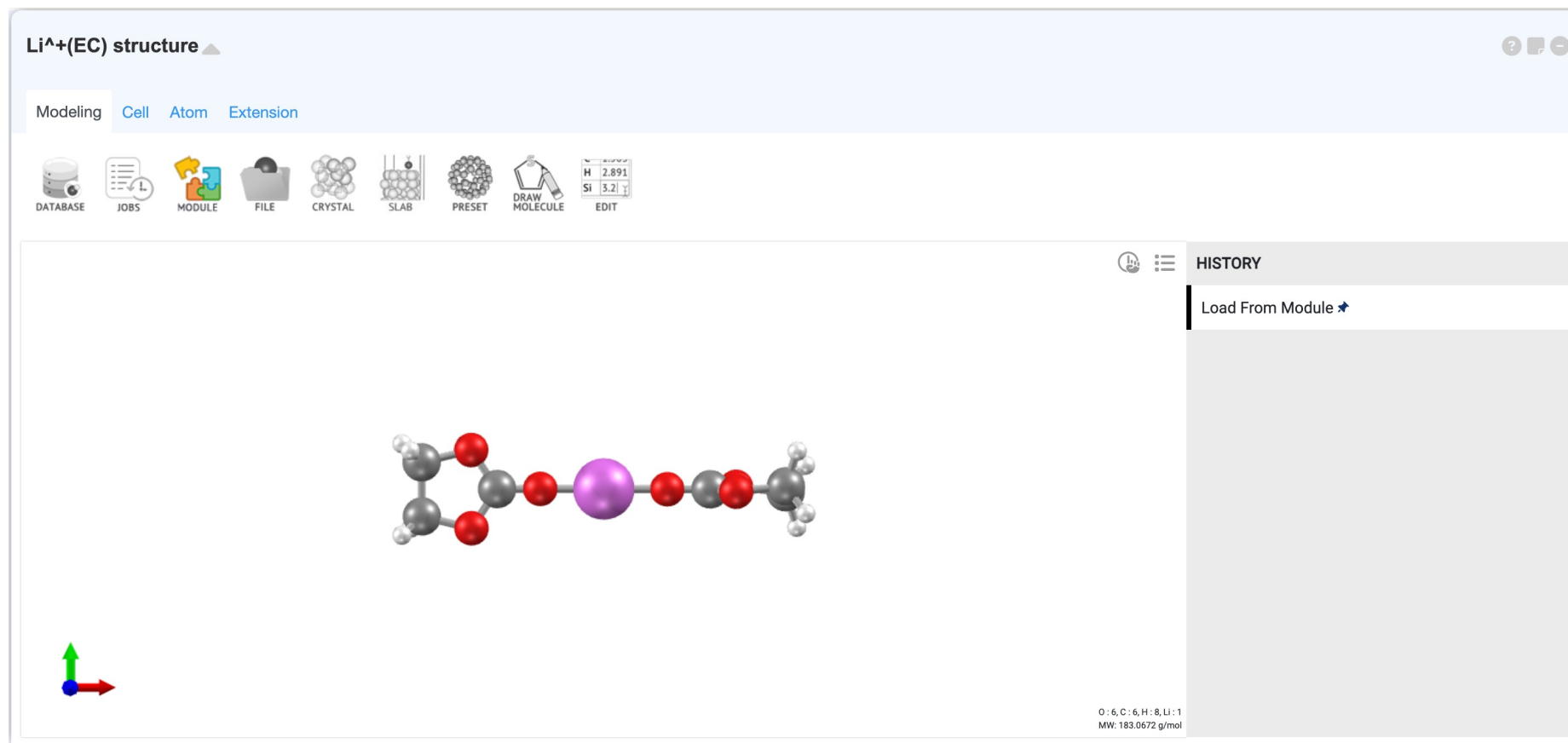
[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

4. Build binding structures



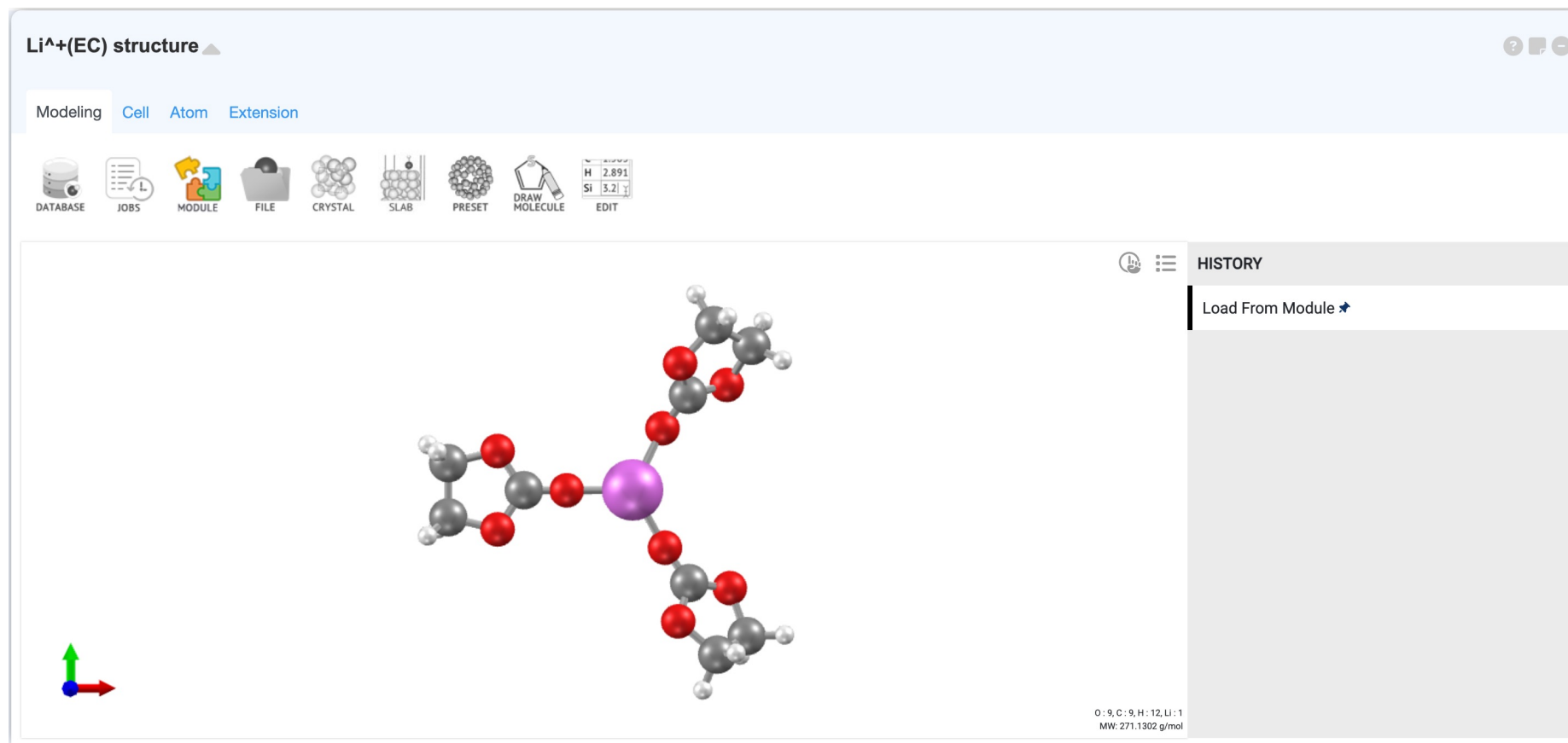
[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

4. Build binding structures



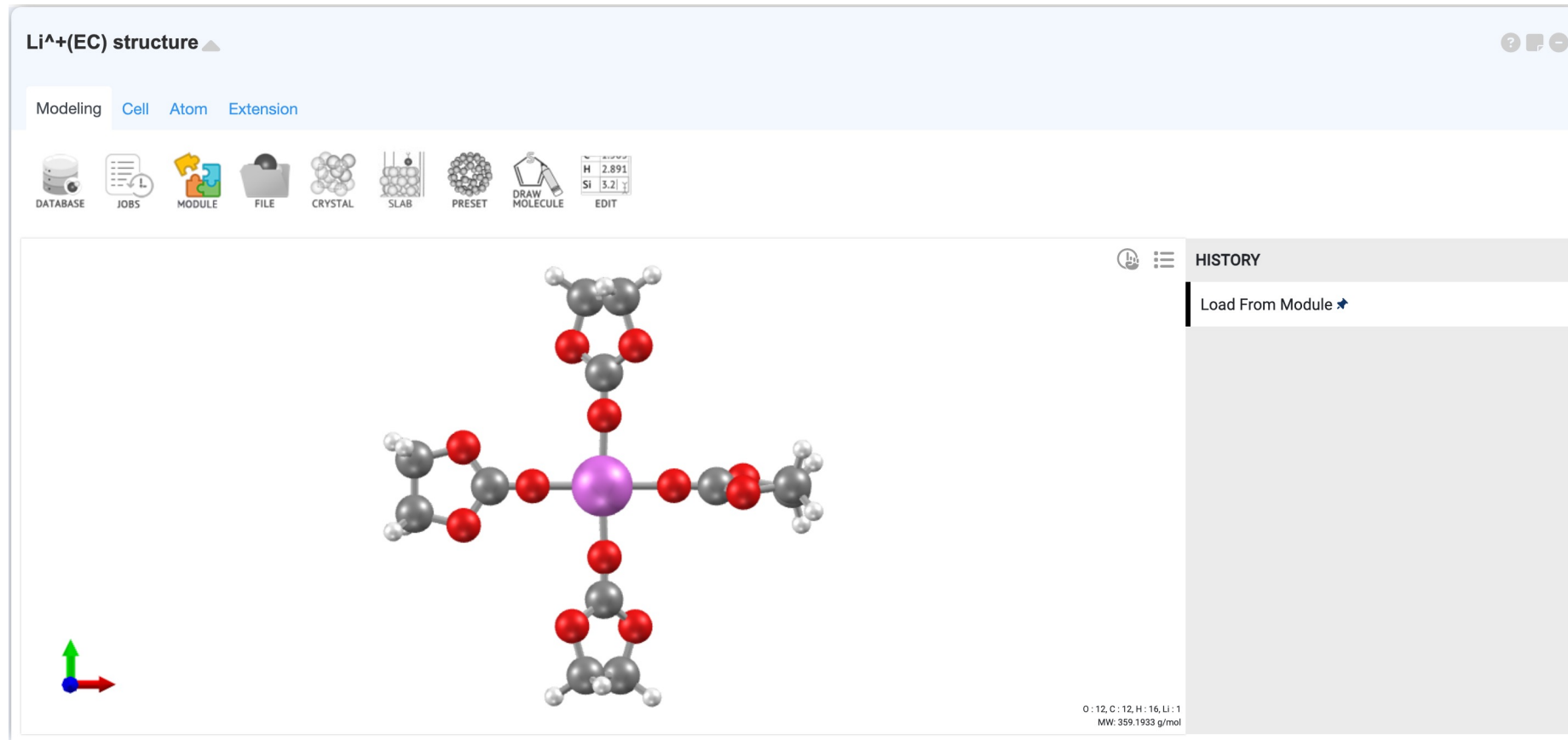
[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

4. Build binding structures



[Lab.5] Interaction of lithium ion with Ethylene Carbonates

4. Build binding structures



[Lab.5] Interaction of lithium ion with Ethylene Carbonates

5. Calculate atomic structure optimization (Li-ion with *n*EC molecules)

GAMESS   

Gameess.x

Result

Fix atoms ▼

Scripting Option

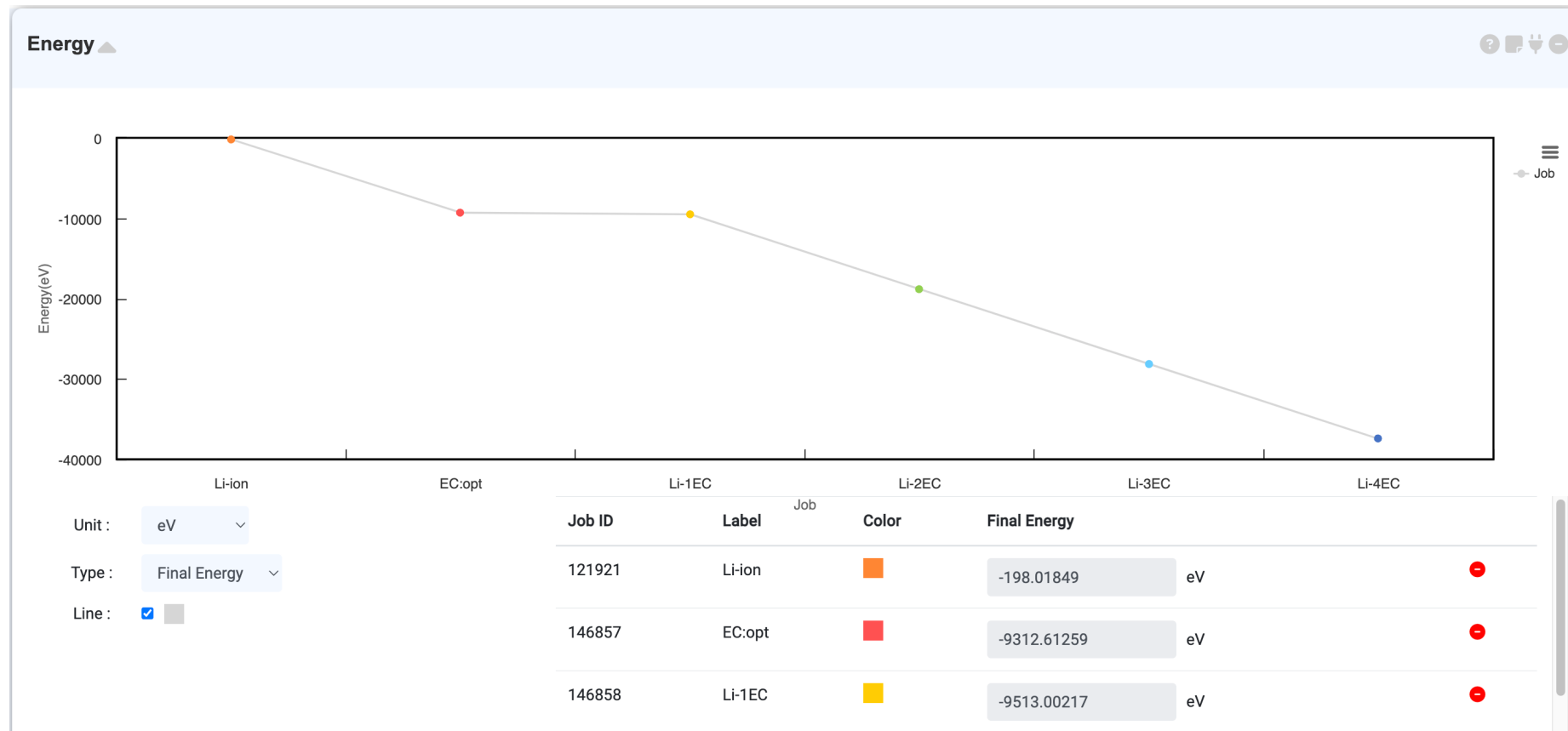
General ▼

\$CONTROL		\$SCF		\$GUESS	
SCF Type	RHF ▼	DIRSCF	.TRUE. ▼	Initial Orbital	Huckel
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TDDFT	NONE ▼			METHOD	SEMINUM ▼
Charge	1			Temperature (K)	298.15
Multiplicity	1				
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\$STATPT					
Convergence tolerance (Hartree/Bohr)	0.0001				
Max iteration step	500				
Hessian matrix	GUESS ▼				
IR frequency	.TRUE. ▼				

[Lab.5] Interaction of lithium ion with Ethylene Carbonates

6. Calculate binding energies

$$E_{adsorption} = -[E_{adsorption\ structure} - (E_{Li-ion} + n_{EC\ molecules} \times E_{EC\ molecules})] / n_{EC\ molecules}$$



[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

6. Calculate binding energies

$$E_{adsorption} = -[E_{adsorption\ structure} - (E_{Li-ion} + n_{EC\ molecules} \times E_{EC\ molecules})]/n_{EC\ molecules}$$

Memo ▲

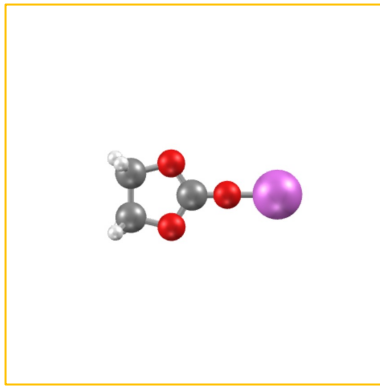
Note Sheet

	A ▼	B ▼	C ▼	D ▼	E ▼	F ▼	G ▼	H ▼	I ▼	J ▼	K ▼	L ▼
1												
2		Total Energy (eV)	Adsorption Energy (eV)		C-O bond distance (Ang)	Li-C bond distance (Ang)						
3	Li	-198.01845										
4	EC	-9312.6125			1.195							
5	Li+(EC)1	-9513.0025	-2.3710895		1.230	1.751						
6	Li+(EC)2	-18827.465	-2.1091500		1.222	1.797, 1.80						
7	Li+(EC)3	-28141.245	-1.7956500		1.214	1.856~1.80						
8	Li+(EC)4	-37454.575	=(B8-(B3+4*B4))/4		1.205	1.977~2.0						
9												
10												

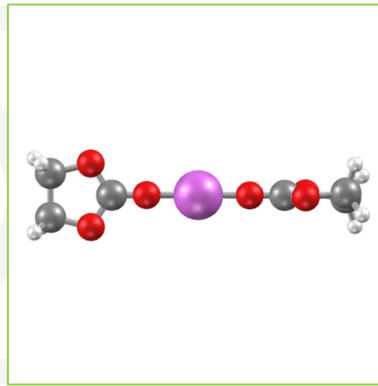
[Lab.5] Interaction of lithium Ion with Ethylene Carbonates

6. Calculate binding energies per molecules

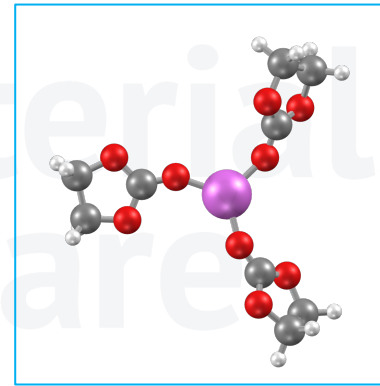
$$E_{adsorption} = -[E_{adsorption\ structure} - (E_{Li-ion} + n_{EC\ molecules} \times E_{EC\ molecules})]/n_{EC\ molecules}$$



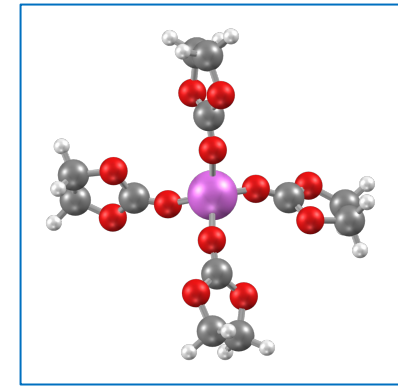
$$E_{adsorption/EC} = 2.37\ eV$$



$$2.11\ eV$$



$$1.80\ eV$$



$$1.53\ eV$$

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Thank you

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